Christian Kim

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EDUCATION

University of California – Riverside Expected June 2026
Bachelor of Science in Electrical Engineering - Control and Robotics GPA: 3.46/4.00
Relevant Coursework: Modeling and Simulation of Dynamic Systems, Programming Practical Robotics, Intro. to Communication Systems

EXPERIENCE

Highlander Space Program – *Liquid Propulsion Rocketry Club* **Riverside, CA**
Payload Lead Engineer / PCB Librarian May 2025 – Present

- Directed the advancement of a mock C.U.T.I.E. CubeSat payload, deployed at apogee, integrating a hyperspectral camera with the CITRUS onboard computer and downlink telemetry to a ground station.
- Designed a CUTIE CubeSat **Battery Card** for a 4S Li-ion pack, integrating BQ77915 front protection, BQ34Z100 **fuel gauging**, and a BQ25713 buck-boost charger/**power-path** controller, with filtered voltage/current/NTC sensing, passive cell balancing, and MOSFET-based charge/discharge isolation to deliver protected 5 V and 3.3 V rails to the CITRUS onboard computer and RF telemetry subsystem.
- Executed a high-speed **Ethernet** PCB (100BASE-TX, 100Ω diff-pairs, DP83848 PHY, magnetics-RJ45) for the pad system; enacted **impedance-controlled** routing, ESD protection, and STM32F7 MAC integration for DAQ networking.

Aviat'R – *Unmanned Autonomous Drone Club* **Riverside, CA**
Avionics Lead / Co-Founder March 2024 – October 2025

- Implemented the avionics power architecture around a 12S, 120A distribution board, ensuring redundant regulation, protected power paths, and reliable telemetry measurement consistent with IPC safety and design practices.
- Configured and validated RFD900x-US radios for long range (>40 km) command and telemetry, optimizing **RF parameters** to meet SUAS mission requirements and maintaining FCC-compliant operation across 902-928 MHz band.

Trustworthy Autonomous Systems Laboratory **Riverside, CA**
Undergraduate Research Assistant October 2024 – January 2025

- Integrated and optimized various module datasets and real-time evaluations to support robust performance on modern GPU architectures, achieving 28 fps on an RTX4090 using Gaussian Splatting.
- Built and integrated the ROSmaster R2 robotic platform (**sensors, drivers, firmware**) for sim-to-real transfer experiments led by PhD researchers in autonomous navigation.

Highlander Racing – *Formula SAE Electric Racing* **Riverside, CA**
Associate/Electrical Systems Engineer August 2023 – December 2024

- Oversaw development of the **Battery Management System** PCB for the team's 400 V pack, integrating an **STM32** with **BQ76PL-Q1** stacked cell monitor, DC-DC power stages, galvanic isolation, and **CAN** bus.
- Designed a front Sensor Hub PCB around an STM32F405, integrating throttle/brake/steering/shock pots, pressure sensors, and thermistors with band-tuned RC filters (6–515 Hz), dual-CAN transceivers with 60Ω split termination + EMI/ESD protection, 3.3 V power regulation, oscillator/boot/reset circuitry, and sealed Deutsch/Ampseal connectors for rugged vehicle telemetry.
- Executed electrical validation, characterizing isolation integrity, contactor coil behavior, **CAN** signal integrity, inverter-induced **EMI** susceptibility, current-sensor accuracy, and thermal performance under transient race-load conditions.
- Implemented **CAN** communication protocol between sensor hubs, VCU, and Inverter; revised PCB layouts to meet IPC-2221B / FSAE standards and produced Gerbers for manufacturing.

PROJECTS

Custom BMS & Electronic Load – Senior Design Project using Altium October 2025 - Present

- Achieved and built a **300W** programmable **electronic load** using multiple **MOSFET** linear power stages (IRFP250N), precision shunt networks, TL431-stabilized analog loops, and an STM32F3 ADC/DAC path; implemented thermal derating, fan-PWM control, iso-trigger input, and kelvin-sensed voltage readback to achieve current regulation across 0-20A.
- Designed a 10-cell (36–42 V) Li-ion **BMS** with BQ76930, 20 mΩ/7 W shunt, 55 V high-side **MOSFET** breakers, and 50–60 A **OC/SC protection**; implemented passive cell balancing, NTC-based thermal cutoff (~65 °C), and a regulated 12 V → 3V3 power tree for deterministic high-voltage pack control.

FPGA-based SDR – RF project (schematic capture done, layout underway). May 2025 - Present

- Designed a HackRF-style SDR architecture in Altium using the **AD9364 RF transceiver**, custom power tree (+1.0V/1.8V/1.35V/3.3V), clock fanout, and high-speed differential data routing into an **Artix-7 FPGA** mezzanine interface
- Implemented a Python-based reference model for real-time IQ capture and digital down-conversion (NCO mixing, LPF, decimation), used to validate the DSP chain prior to STM32 and FPGA implementation

SKILLS AND INTERESTS

Software: Altium Designer, LTSpice, Git, GNU Radio, Vivado/Vitis, OnShape
Programming: Python, MATLAB, C++
Hardware: STM32, USB, CAN, UART, I2C, SPI, MCUs, oscilloscope, power supplies
Manufacturing: Soldering, manual PCB assembly, 3D printing